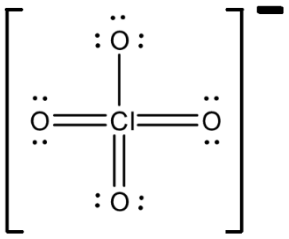
Chemistry 3A – Fall 2026 **Version B**

Sections 25810-25811

Midterm Examination #2 6 October 2026

Select **BEST** response for question. The point values for questions is now changed based on the difficulty and effort of solving the question. Point values for M/C are now in brackets. All T/F questions are 1 pt

The image at right shows the Lewis structure for the polyatomic ion, the perchlorate ion (ClO_4^-)
The **<u>next four questions</u>** relate to it.



1. What are the total number of electrons (valence electrons of all atoms and any electrons creating the ion) for the molecule? [3]
   a) 7 b) 31 c) 32 d) 62
2. How many valence electrons does **chlorine (Cl)** bring to the perchlorate ion? [2]
   a) 2 b) 6 c) 7 d) 8
3. Which of the following statements is TRUE about Lewis structures? [2]
   a) The central atom **chlorine** has ***seven*** (**7**) **lone** (**nonbonding**) electron pairs
   b) In order to get a -1 net charge an additional electron had to be added to the valence electrons of all the atoms to make the molecule an ion
   c) The molecule has a net charge of -3
   d) Three oxygen (O) atoms ***each*** have three (3) lone (nonbonding) pairs
4. Which of the following is a FALSE statement about the perchlorate ion shown above? [2]
   a) Chlorine forms three double bonds and one single bond with oxygen atoms in this resonance structure
   b) This polyatomic ion has a **linear *molecular geometry***
   c) The central chlorine (Cl) atom has four (4) **electron groups** around it
   d) The perchlorate ion carries a net charge of –1 (minus 1)

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5. What is **electronegativity**? [2]
   a) the number of lone pairs of electrons an atom has
   b) it is related to the repulsion theory that explains large and small nuclei
   c) potential of an atom to pull or attract electrons to itself in a chemical bond
   d) It explains why electrons emit electromagnetic waves without changing shell energy levels
6. Which of these types of matter will have *definite volume* and a *definite shape*? [2]
   a) gas b) liquid c) solid d) plasma
7. What is percentage of the mass of calcium chloride dihydrate $CaCl_2 . 2\ H_2O$ is ***NOT*** water (H_2O), that is, what percentage is just $CaCl_2$? [4]
   a) 85.98% b) 75.49% c) 63.92% d) 24.51%
8. There are **6.022×10^{23} atoms** of element **sodium** (**Na**). How many **moles** of **Na** do you have? [3]
   a) 6.022×10^{23} g b) 6.022×10^{23} mol c) 23.99 mol d) 1 mol

9. A 10.0 g piece of zinc (Zn) metal is cooled from 30.0°C to 10.0°C. Energy transfer was –80.0 J (joule) (if less than zero, that is heat energy released). What is specific heat capacity of zinc? [4]

a) –0.400 J/(g °C) b) 0.400 J/(g °C) c) 2.50 J/(g °C) d) 0.200 J/(g °C)

10. What is the best choice for what is TRUE about the characteristics of a Type II ion? [2]

a) it is a polyatomic anion composed of nonmetal elements
b) it is an element that will ionize only to one charge state, such as +1 or +2 or +3 or more, but only one of those
c) the element's name in a compound requires the use of a roman numeral (the Stock system)
d) it only applies to Group 2 elements, which is why it is called a Type II ion

11. In STEPS detailed in lecture for drawing a Lewis structure for the carbonate ion CO_3^{2-} ion, what step comes AFTER the total of all electrons (valence and adjusting for the ionic charge on the polyatomic ion) are obtained? [2]

a) Look at all the valence electrons of all atoms and adjust for the charge of the ion
b) Determine the central atom and arrange the connecting or terminal atoms around it
c) Add hydrogen (H) atoms to the structure
d) Make sure the central atom has an octet by creating double and triple bonds with other atoms

12. Which statement best explains why liquid mercury forms a convex (curved upward) meniscus in a glass tube, whereas water forms a concave meniscus? [2]

a) Water has stronger cohesive forces than adhesive forces with glass.
b) Mercury has stronger adhesive forces with glass than cohesive forces among its atoms.
c) Water's adhesive forces to glass are stronger than its internal cohesive forces, while mercury's internal cohesive forces exceed its adhesive forces to glass.
d) Mercury's density causes it to push away from glass walls regardless of surface forces.

13. The molecule for water (H_2O) has two terminal H atoms around the central O atom, which has two nonbonding (lone) pairs of electrons. What is the molecular geometry of H_2O? [2]

a) bent b) trigonal planar c) trigonal pyramidal d) tetrahedral

14. Sodium (Na) and chlorine (Cl) show a somewhat large difference in their electronegativity values. What kind of chemical bond are Na and Cl likely to form as a compound? [2]

a) ionic b) metallic c) (nonpolar) covalent d) polar covalent

15. Solids of matter such as ice (H_2O solid) and iodine (I_2 solid) are classified as what type of solid? [2]

a) ionic b) covalent network c) metallic d) molecular

16. What happens to the temperature of a pure substance while it is undergoing melting at its normal melting point on a heating curve? [2]

a) The temperature increases steadily as heat is added)
b) The temperature decreases temporarily due to heat absorption.
c) The temperature remains constant until the phase change is complete.
d) The temperature fluctuates depending on the rate of heating

17. When atoms like rubidium and chlorine have the largest differences in electronegativity, what kind of bond will they form? [2]

a) ionic b) metallic c) polar covalent d) covalent-ionic

18. What is the molar mass (units = g/mol) of ammonium carbonate, $(NH_4)_2CO_3$? [4]

Atomic masses: N = 14.01 g/mol, H = 1.008 g/mol, C = 12.01 g/mol, O = 16.00 g/mol

a) 78.05 b) 96.09 c) 112.09 d)144.18

19. You have **100 g** of an element whose molar mass is **50 g/mol**. How many moles do you have of the element? [3]

a) 50 mol b) 0.5 mol c) 2 mol d) 5,000 mol

20. Given the **electronegativity** (EN) values: H = 2.1, C = 2.5, O = 3.5, Na = 0.9, Cl = 3.0. Which bond is classified as **polar covalent**? (*EN difference* criteria: nonpolar <0.5, polar 0.5--2.0, ionic >2.0) [2]

a) C–H bond b) O–H bond c) Na–Cl bond d) C–C bond

21. What kind of **intermolecular force** (IMF) would be expected for a molecule like chloromethane (CH_3Cl), which has slight molecular polarity, and DISTINCT from the IMF seen in HF or H_2O? [2]

a) London dispersion forces b) hydrogen bonding
c) covalent-ionic attraction d) dipole-dipole interaction

22. What is the correct systematic name for the compound with type II metal cation **$CoBr_2$**? [2]

a) cobalt bromide b) cobalt(I) bromide c) cobalt(II) bromide d) cobalt(II) dibromide

23. If you have a **Mg^{2+}** ion and a **PO_4^{3-}** ion, what compound will it form? Apply the criss-cross method! [2]

a) $Mg_2(PO_4)_3$ b) $MgPO_4$ c) $Mg_3(PO_4)_2$ d) $Mg_3(PO_3)_4$

24. Which molecular formula correctly shows the compound **carbon tetraiodide**? [2]

a) CI b) C_4I c) C_4I_4 d) CI_4

25. Which of the following represents a change in enthalpy in a substance that changes from the solid phase to the gas phase? [2]

a) ΔH_{vap} b) ΔH_{fus} c) $\Delta H_{endoexo}$ d) ΔH_{sub}

The following statements are either True or False. Select **choice A** for **true**, or **choice B** for **false**

26. Sulfur dioxide (SO_2) is a molecular compound.
    a) true b) false
27. Valence Shell Electron Pair Repulsion (VSEPR) theory explains why the three atom molecule H–O–H (water) with two lone pairs on oxygen has a bent molecular geometry and three atom molecule O=C=O (carbon dioxide) with no lone pairs on carbon has a linear molecular geometry
    a) true b) false
28. A double bond between two carbon atoms is shorter and stronger than a single bond between two carbon atoms.
    a) true b) false
29. The boiling of water to steam is an endothermic process
    a) true b) false
30. Aluminum is a Group 13 element
    a) true b) false
31. 1 gram of sodium (Na) atoms and 1 gram of chlorine (Cl) atoms will have equal numbers of Na and Cl atoms
    a) true b) false
32. Nitrogen has a molar mass of 9.012 g/mol
    a) true b) false
33. London dispersion forces are the weakest of intermolecular forces, where dipole-dipole interactions and hydrogen bonding are the other forces.
    a) true b) false
34. Boiling is the process of matter that is liquid into matter that is a gas.
    a) true b) false
35. There are more molecules in 1.0 mol of SO_2 than in 0.50 mol of CO_2
    a) true b) false

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